Google Cloud H2S

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

Participant Details

a. Participant Name: Syed Naazim Hussain
b. Problem Statement: Build and deploy AI agents using Gemini, ADK, and Cloud Run.

Brief about the idea

Project Hermes: The Autonomous Content Factory

- Hermes is a production-grade multi-agent swarm that automates the entire lifecycle of YouTube Shorts.
- It transforms a high-level goal (e.g., "Create a video about sustainable tech") into a researched, scripted, produced, and scheduled video, entirely without human intervention.
- **Mission**: To bridge the gap between human creativity and autonomous execution using Agentic Workflows.

Your solution should be able to explain the following:

- How did you approach the common problem statement and translate it into a working solution using ADK, MCP/AlloyDB AI while ensuring Hackathon's core requirements are met?
- What real-world problem does your solution address, and what practical impact does it create for users?
- What is the core approach or workflow behind your solution?

- **Approach**: We moved away from "Linear Pipelines" to "Collaborative Swarms." Using the **Google Agent Development Kit (ADK)**, we built a system where agents communicate via a shared state, allowing for recursive research and self-correction during scriptwriting and production.
- **Real-World Problem**: Manual content creation is the #1 bottleneck for small creators and brands. Research, scripting, and editing take 6–8 hours per video.
- **Practical Impact**: Hermes reduces this to **under 10 minutes** of autonomous background processing, enabling creators to scale their voice without the burnout.
- **Workflow**: Coordinator (Orchestrator) → Research/Ideas (Discovery) → ScriptWriter (Creative) → Producer (Media Gen) → Scheduler (Distribution).

Opportunities

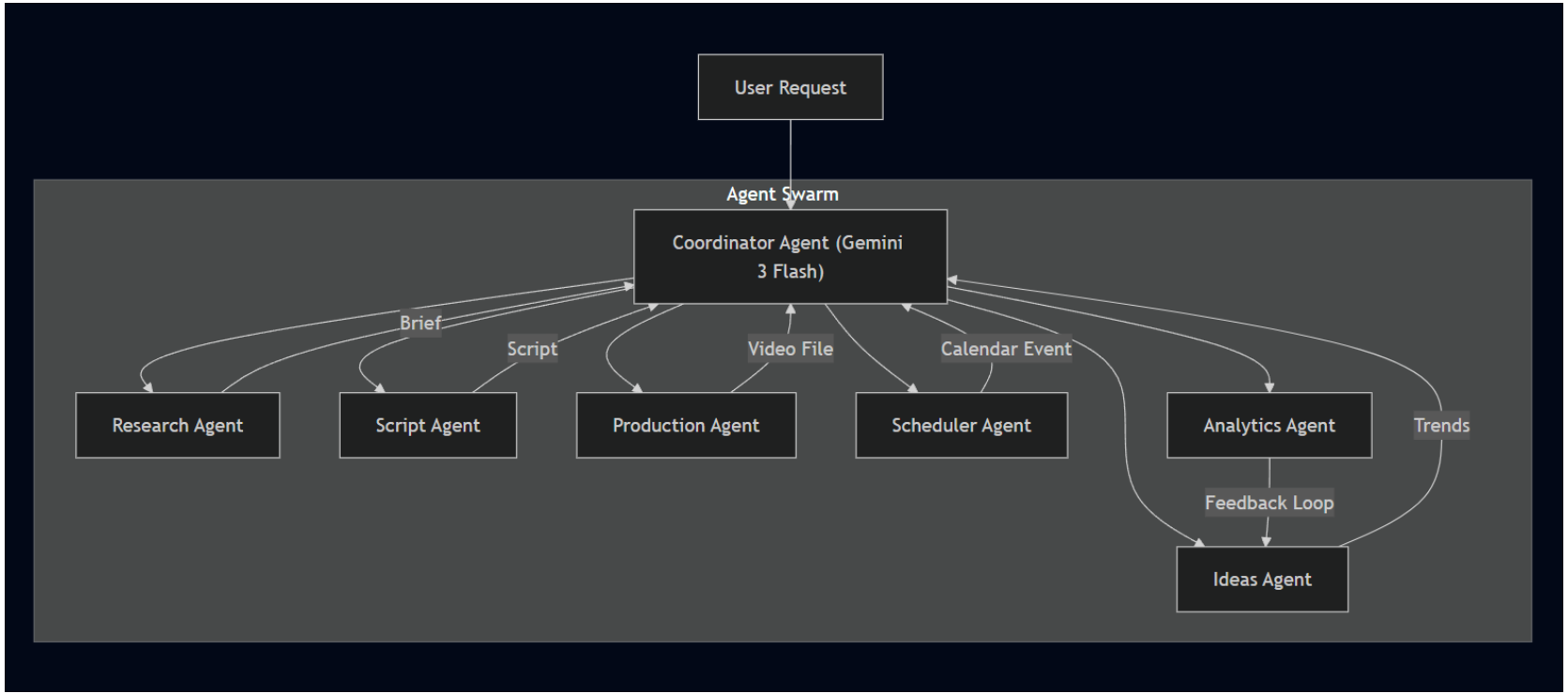
- How different is it from any of the other existing ideas?
- USP of the proposed solution

- Unlike existing "AI Editors" (which require human inputs at every step), Hermes is **fully autonomous**. It finds the trends for you using live data, rather than just reacting to your text.
- **Closed-Loop Analytics**: Most tools stop at "Publish." Hermes includes a daily feedback loop that analyzes your channel's metrics to find what's working and automatically feeds those insights back into the "Ideas Agent" for next week’s content.
- **USP**: **Zero-Touch Video Production**. From Trend to YouTube Studio in one click.

List of features offered by the solution

- **Agentic Orchestration:** Multi-agent task decomposition using Gemini 3 Flash.
- **Live Trend Integration:** Real-time discovery via HackerNews and Google Trends.
- **Recursive Research:** Fact-finding missions using Tavily and Firecrawl.
- **Native TTS & Visuals:** Automated voiceovers and Imagen-powered scene generation.
- **Smart Scheduling:** Integration with Google Calendar for optimal posting windows.
- **Analytics Flywheel:** Automatic topic scoring based on YouTube performance.

Architecture diagram of the proposed solution:



Technologies/Google Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

The Core Stack

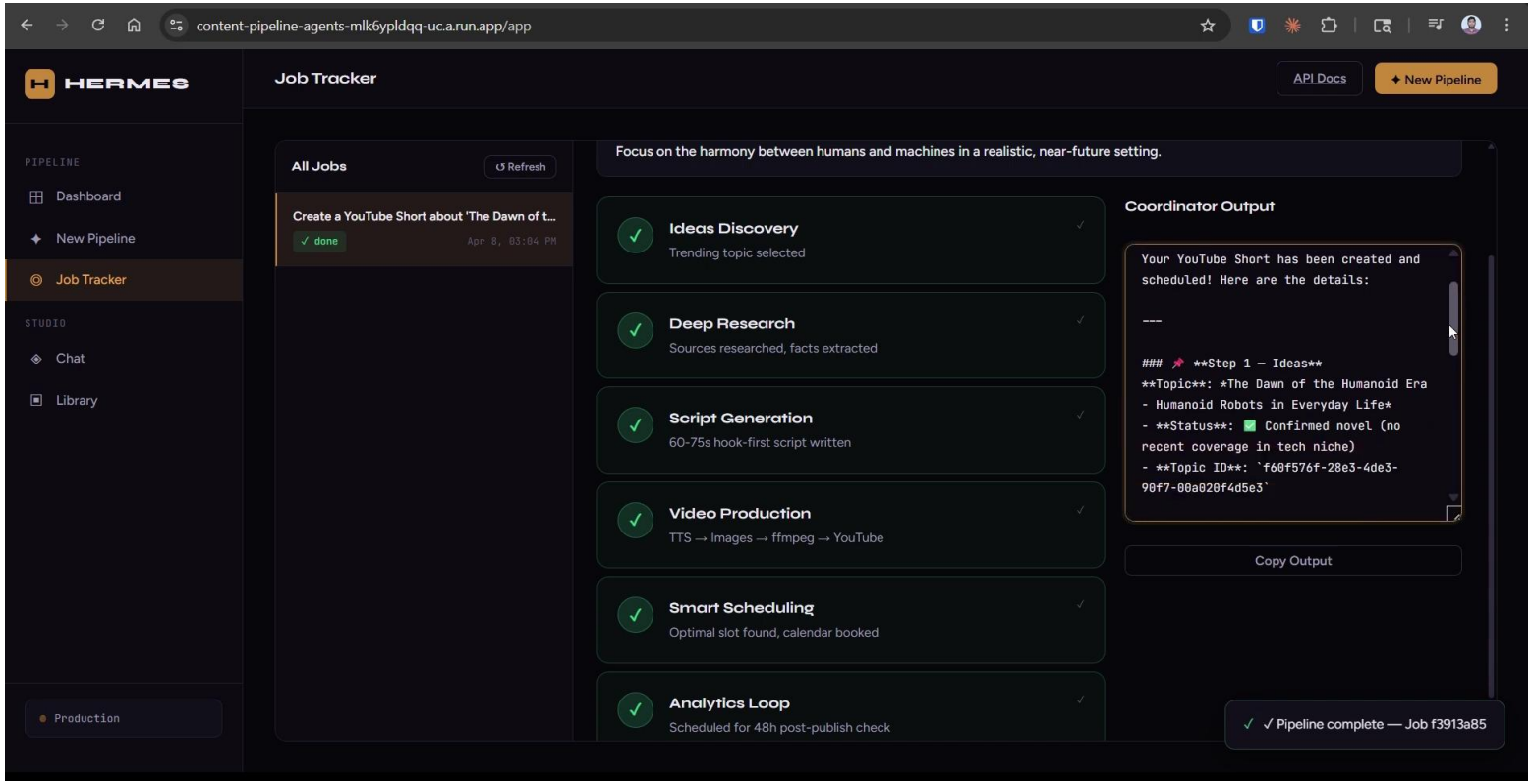
- **Gemini 3 Flash**: The "Central Brain." We chose Flash for its **high-throughput** and **1M+ context window**, which is critical for ingesting hundreds of search results into a single coherent research brief.
- **Google ADK (Agent Development Kit)**: Our orchestration engine. It allows us to build complex, tool-using agents that can self-correct and collaborate without hard-coded logic.
- **Google Cloud Run (Services + Jobs)**:
- **Services**: Handle the API and Dashboard logic.
- **Jobs**: Specifically used for **Media Production (ffmpeg/Imagen)**. This allows us to scale heavy processing tasks independently of the web server.
- **Cloud Firestore**: Our global "Shared Memory." It stores real-time agent state, allowing for persistent job monitoring even if a container restarts.
- **Secret Manager**: Ensures zero-trust security for YouTube OAuth tokens and API keys.

Technologies/Google Services used in the solution

(Why did you choose your specific AI stack and system design, and how does it support scalability and real-world deployment?)

- **Why this Design?**
- **Scalability**: By moving the "Production Agent" to a **Cloud Run Job**, we can generate 1,000 videos simultaneously. Each job gets its own dedicated 2-CPU/2GB environment, ensuring zero performance interference.
- **Economic Efficiency**: A purely serverless architecture means **Zero cost when idle**. We only pay for the exact seconds the agents are "working."
- **Real-World Reliability**:
- **Fault Tolerance**: If a research agent fails due to an external API error, the ADK Runner can automatically retry the tool-call.
- **Global Deployment**: Containerization allows us to deploy Hermes near the creator, reducing latency for uploading massive 4K video files to YouTube.

Snapshots of the prototype



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Thank you